Physics potential : top-FCNC

LHeC@1.2 TeV & LHμC@3.7 TeV

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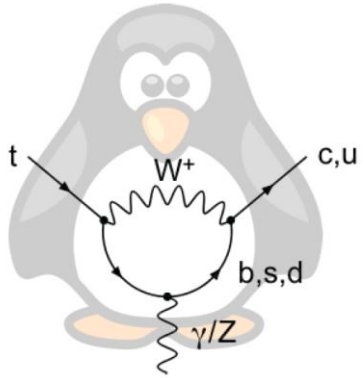
Outline: LHeC@1.2 TeV & LHμC@3.7 TeV

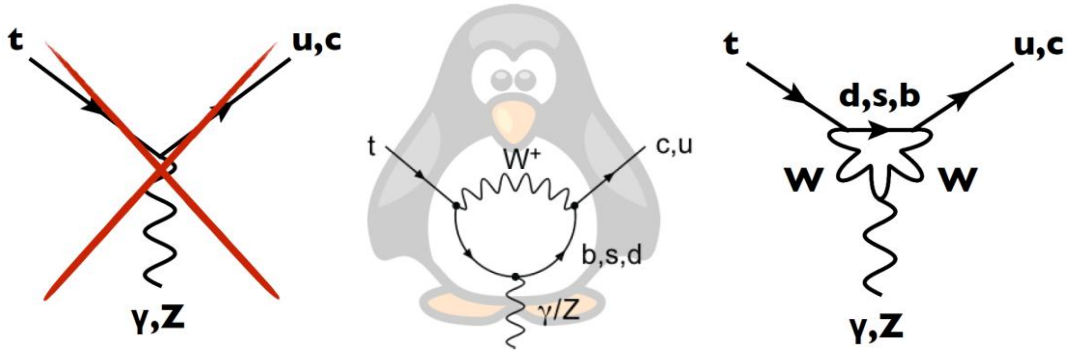
- Top-quark flavor-changing neutral currents : top-FCNCs
- FCNC production of single top : top-FCNCs (tqZ, $tq\gamma$)
- Top-FCNCs : Semi-leptonic decay channel

$e^- p \to e^- t$ (via FCNCs tqZ, $tq\gamma$) $t \to W^+ b, \quad W^+ \to \ell^+ \nu_\ell \; (\ell = e, \mu)$



The top quark FCNCs @ LHeC@1.2 TeV & LHμC@3.7 TeV

- The Flavor-changing neutral current (FCNC) process is forbidden at tree level in the SM and is highly suppressed in one-loop corrections by the Glashow-Iliopoulos-Maiani mechanism.
- The FCNC interactions of the tqX vertex (where *q* represents up and charm quarks) are key processes to search for New Physics.



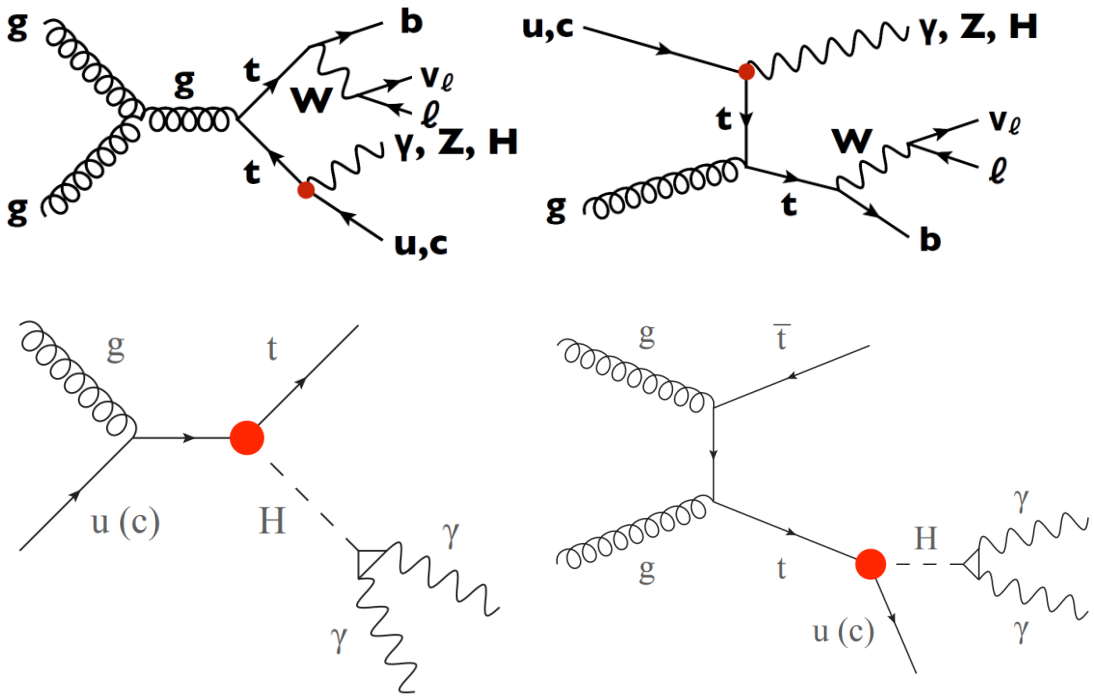
- The SM expected top quark FCNC transition is far from the sensitivity of the current experiments.
- The FCNC decay rates can be significantly enhanced in some New Physics models.
- The observation of a top quark FCNC interaction would signal the existence of New Physics.

	SM	QS	2HDM	FC 2HDM	MSSM	$\not{R}$ SUSY
$t \to uZ$	8×10^{-17}	1.1×10^{-4}	–	–	2×10^{-6}	3×10^{-5}
$t \to u\gamma$	3.7×10^{-16}	7.5×10^{-9}	–	–	2×10^{-6}	1×10^{-6}
$t \to ug$	3.7×10^{-14}	1.5×10^{-7}	–	–	8×10^{-5}	2×10^{-4}
$t \to uH$	2×10^{-17}	4.1×10^{-5}	5.5×10^{-6}	–	10^{-5}	$\sim 10^{-6}$
$t \to cZ$	1×10^{-14}	1.1×10^{-4}	$\sim 10^{-7}$	$\sim 10^{-10}$	2×10^{-6}	3×10^{-5}
$t \to c\gamma$	4.6×10^{-14}	7.5×10^{-9}	$\sim 10^{-6}$	$\sim 10^{-9}$	2×10^{-6}	1×10^{-6}
$t \to cg$	4.6×10^{-12}	1.5×10^{-7}	$\sim 10^{-4}$	$\sim 10^{-8}$	8×10^{-5}	2×10^{-4}
$t \to cH$	3×10^{-15}	4.1×10^{-5}	1.5×10^{-3}	$\sim 10^{-5}$	10^{-5}	$\sim 10^{-6}$

Aguilar-Saavedra J.A., Top flavor-changing neutral interactions: Theoretical expectations and experimental detection. *Acta Physica Polonica* B, 35 (2004) 2695.

Summary of ATLAS and CMS results

- Over the years, various experiments have searched for the FCNC processes through the anomalous production or decay of top quark.
- No clear evidence for the presence of top quark FCNC interactions is observed and upper limits are set on the anomalous couplings and top quark branching ratio.



Boson	Best Limit on the BR (95% CL)		
γ	ATLAS 81 fb^{-1}, focus on $\sigma_{t+boson}$	~0.3 x 10^{-4} (up)	~2 x 10^{-4} (charm)
Z	ATLAS 36 fb^{-1}, focus on BR	~2 x 10^{-4} (up)	~2 x 10^{-4} (charm)
Higgs	CMS 137 fb^{-1}, $H \rightarrow \gamma\gamma$	~2 x 10^{-4} (up)	~7 x 10^{-4} (charm)
gluon	CMS Run-1, top+jet production	~0.2 x 10^{-4} (up)	~4 x 10^{-4} (charm)

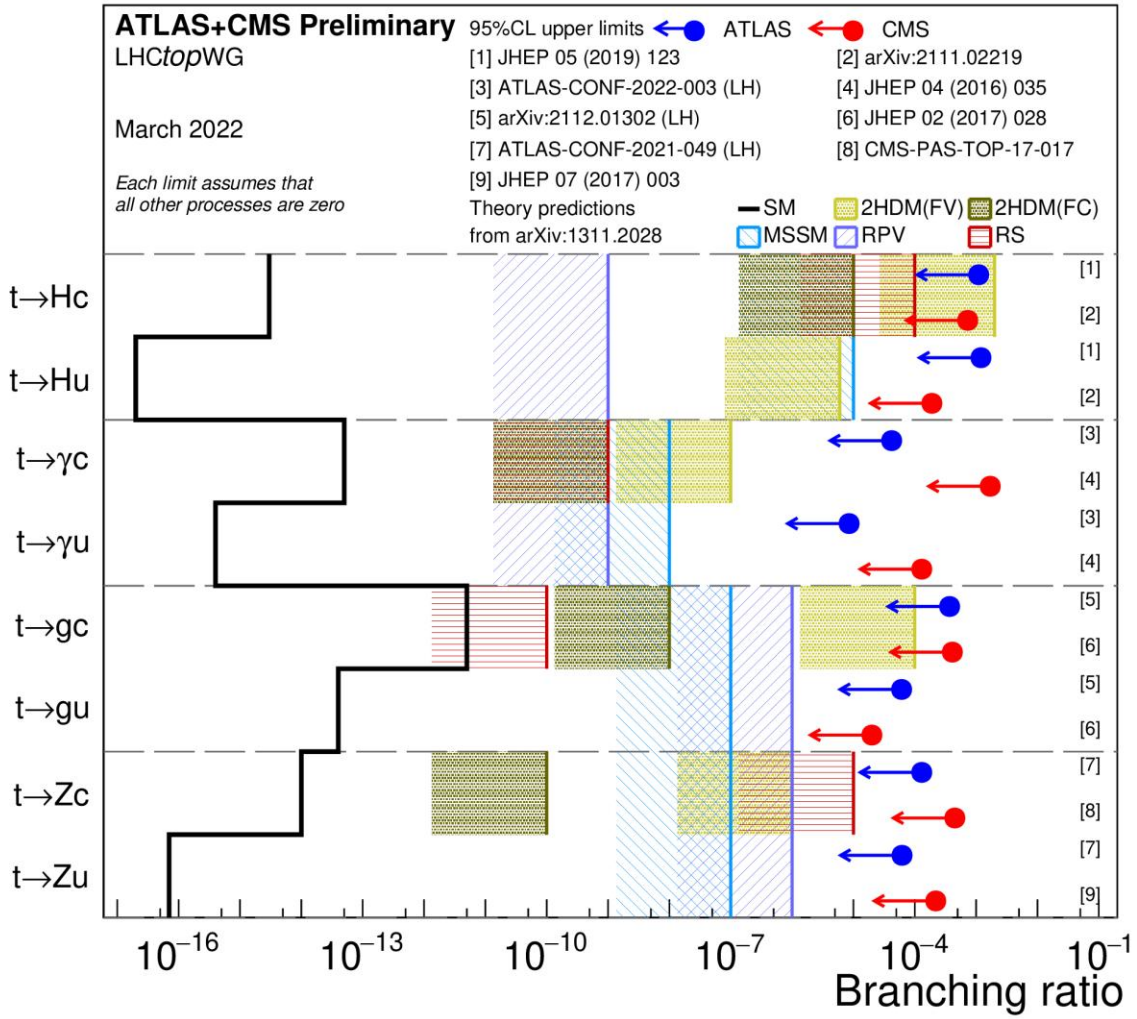
ATLAS Collaboration, Phys. Lett. B 800 (2020) 135082, 1908.08461 [hep-ex].
ATLAS Collaboration, ATLAS-CONF-2022-003 & ATLAS-CONF-2022-013.
ATLAS Collaboration, CMS-TOP-20-007, 2111.02219 [hep-ex].
ATLAS Collaboration, JHEP02 (2022) 169, 2112.09734 [hep-ex].

The SM & beyond the SM & Experimental results

- Several extensions of the SM predict much higher branching ratios up to 10^8 - 10^9 order of magnitude larger than SM predictions.

- Any observation of these rare FCNC transitions would be a clear signal of new physics beyond the SM.

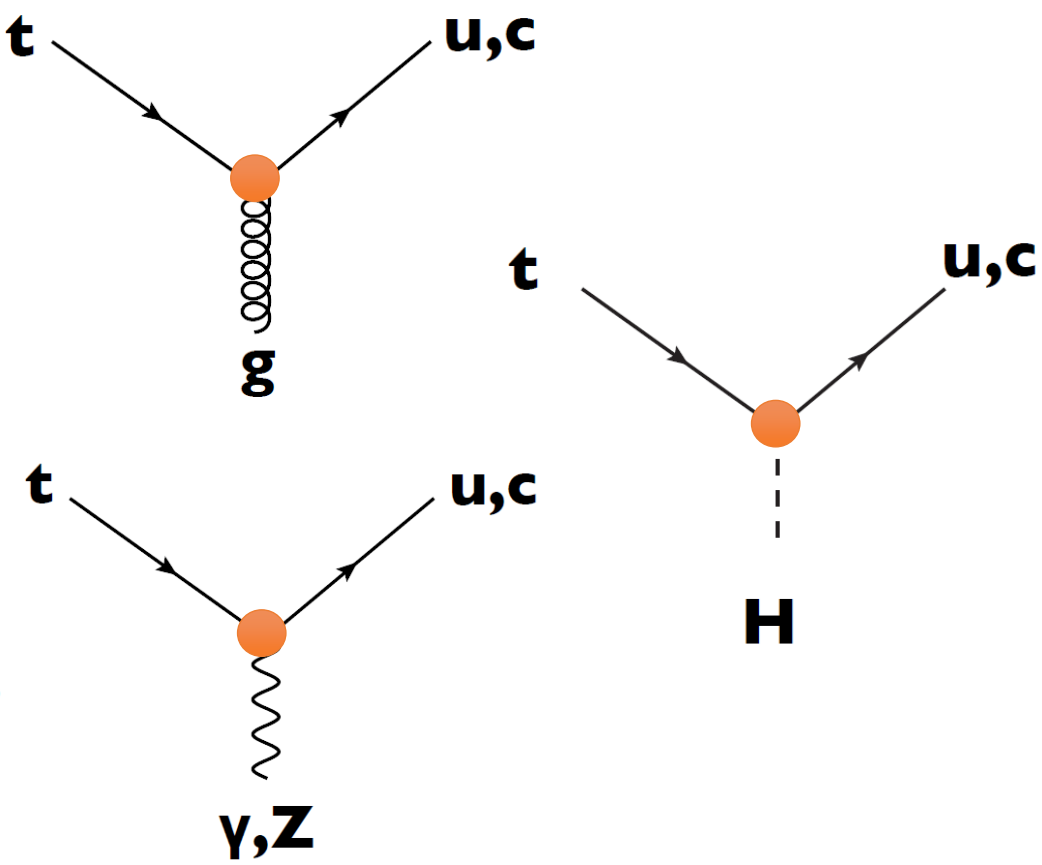
Aguilar-Saavedra J.A., Top flavor-changing neutral interactions: Theoretical expectations and experimental detection. *Acta Physica Polonica* B, 35 (2004) 2695.



The Effective Lagrangian Approach

- The most general effective Lagrangian describing the FCNC interactions can be parametrized as

$$
\begin{aligned}
\mathcal{L}_{\mathrm{FCNC}} = \sum_{q=u,c} \Bigg[& \frac{g_s}{2m_t} \bar{q}\lambda^a \sigma^{\mu\nu} (\zeta^L_{qt} P^L + \zeta^R_{qt} P^R) t G^a_{\mu\nu} \\
& - \frac{1}{\sqrt{2}} \bar{q} (\eta^L_{qt} P^L + \eta^R_{qt} P^R) t H \\
& - \frac{g_W}{2c_W} \bar{q}\gamma^\mu (X^L_{qt} P_L + X^R_{qt} P_R) t Z_\mu \\
& + \frac{g_W}{4c_W m_Z} \bar{q}\sigma^{\mu\nu} (\kappa^L_{qt} P_L + \kappa^R_{qt} P_R) t Z_{\mu\nu} \\
& + \frac{e}{2m_t} \bar{q}\sigma^{\mu\nu} (\lambda^L_{qt} P_L + \lambda^R_{qt} P_R) t A_{\mu\nu} \Bigg] + \mathrm{H.c.},
\end{aligned}
$$



https://feynrules.irmp.ucl.ac.be/wiki/EffectiveModels
https://feynrules.irmp.ucl.ac.be/wiki/GeneralFCNTop

- Top-quark FCNC interactions (tqγ, tqZ, tqH)
- Lepton–hadron colliders: μLHC and FCC-μH

- SMEFT & anomalous FCNC operators (dim-6)
- Event generation & detector modeling for μp

- High-energy PDFs (proton + photon/EPAs)
- MadGraph5_aMC@NLO,
- UFO model implementation, parameter scans

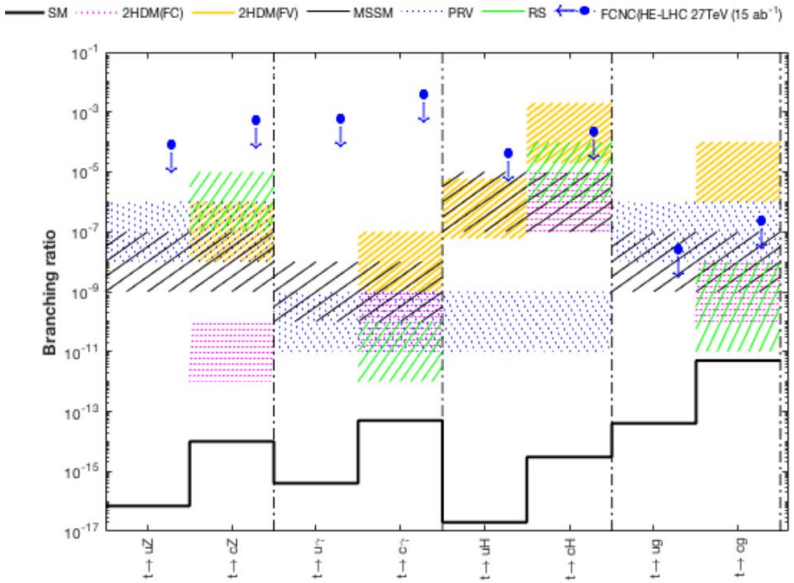
- Signal significance estimates and limit setting

- Top quark FCNC at HE-LHC and FCC-hh

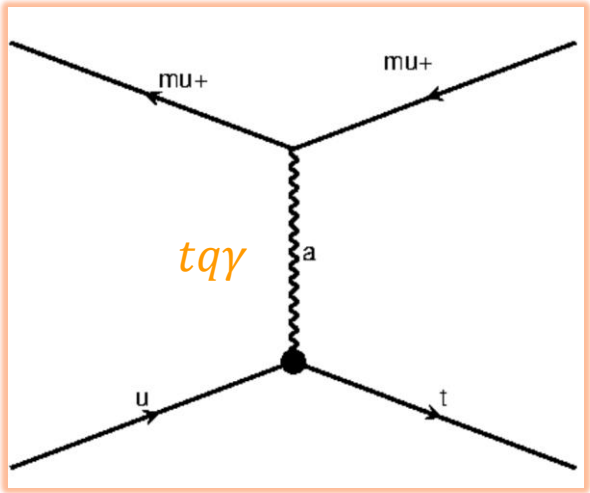
  Hamzeh Khanpour, *Nucl. Phys. B* 958 (2020) 115141

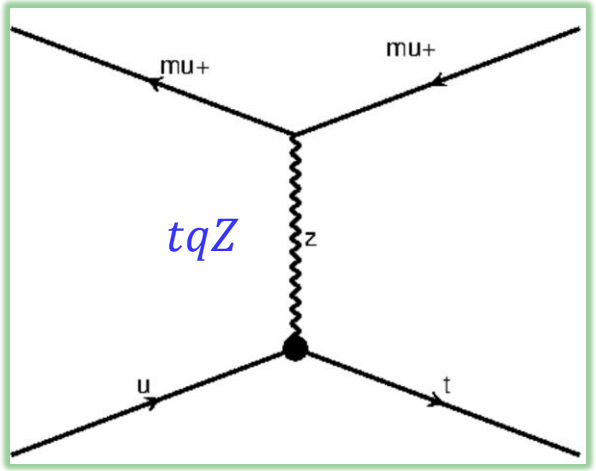
- Top quark FCNC at FCCee

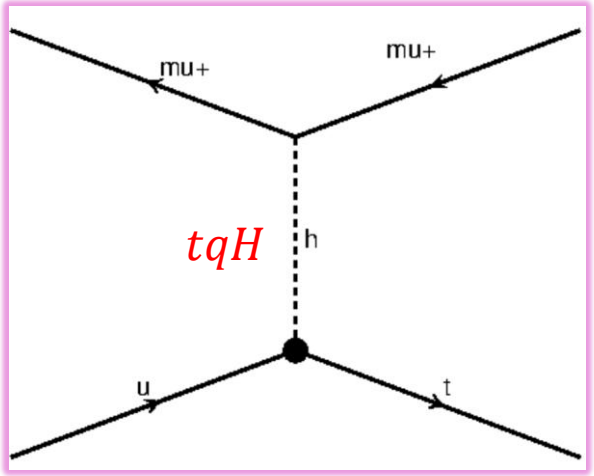
  Hamzeh Khanpour, Phys. Lett. B 775 (2017) 25-31



Top-FCNCs at a lepton-proton collider: $tq\gamma$; tqZ; tqH







$$
\begin{aligned}
\mathcal{L}_{\mathrm{FCNC}} = \sum_{q=u,c} \Bigg[& \frac{g_s}{2m_t} \bar{q}\lambda^a \sigma^{\mu\nu} (\zeta^L_{qt} P^L + \zeta^R_{qt} P^R) t G^a_{\mu\nu} \\
& - \frac{1}{\sqrt{2}} \bar{q} (\eta^L_{qt} P^L + \eta^R_{qt} P^R) t H \\
& - \frac{g_W}{2c_W} \bar{q}\gamma^\mu (X^L_{qt} P_L + X^R_{qt} P_R) t Z_\mu \\
& + \frac{g_W}{4c_W m_Z} \bar{q}\sigma^{\mu\nu} (\kappa^L_{qt} P_L + \kappa^R_{qt} P_R) t Z_{\mu\nu} \\
& + \frac{e}{2m_t} \bar{q}\sigma^{\mu\nu} (\lambda^L_{qt} P_L + \lambda^R_{qt} P_R) t A_{\mu\nu} \Bigg] + \mathrm{H.c.},
\end{aligned}
$$

LeHC@1.2 TeV and LμHC@3.7 TeV : collider configurations and MG settings

- Top-FCNC EFT model: import model topFCNC_UFO

$$p \equiv \{g,\ u,\ c,\ d,\ s,\ b,\ \bar{u},\ \bar{c},\ \bar{d},\ \bar{s},\ \bar{b}\}$$

$$j \equiv \{g,\ u,\ c,\ d,\ s,\ \bar{u},\ \bar{c},\ \bar{d},\ \bar{s},\ b,\ \bar{b}\}$$

- generate p e- > e- t, (t > b lp vl) NP=1

$$p + \mu^+ \longrightarrow \mu^+ + t,$$

LeHC@1.2 TeV

- Top-FCNC EFT model: import model topFCNC_UFO

$$p \equiv \{g,\ u,\ c,\ d,\ s,\ b,\ \bar{u},\ \bar{c},\ \bar{d},\ \bar{s},\ \bar{b}\}$$

$$j \equiv \{g,\ u,\ c,\ d,\ s,\ \bar{u},\ \bar{c},\ \bar{d},\ \bar{s},\ b,\ \bar{b}\}$$

- generate p mu+ > mu+ t, (t > b lp vl) NP=1

$$p + \mu^+ \longrightarrow \mu^+ + t,$$

LμHC@3.7 TeV

https://github.com/hamzeh-khanpour/Top_FCNC_LHeC_LHmuC

Generator-level cuts and event selection

- Generation-level (basic) cuts

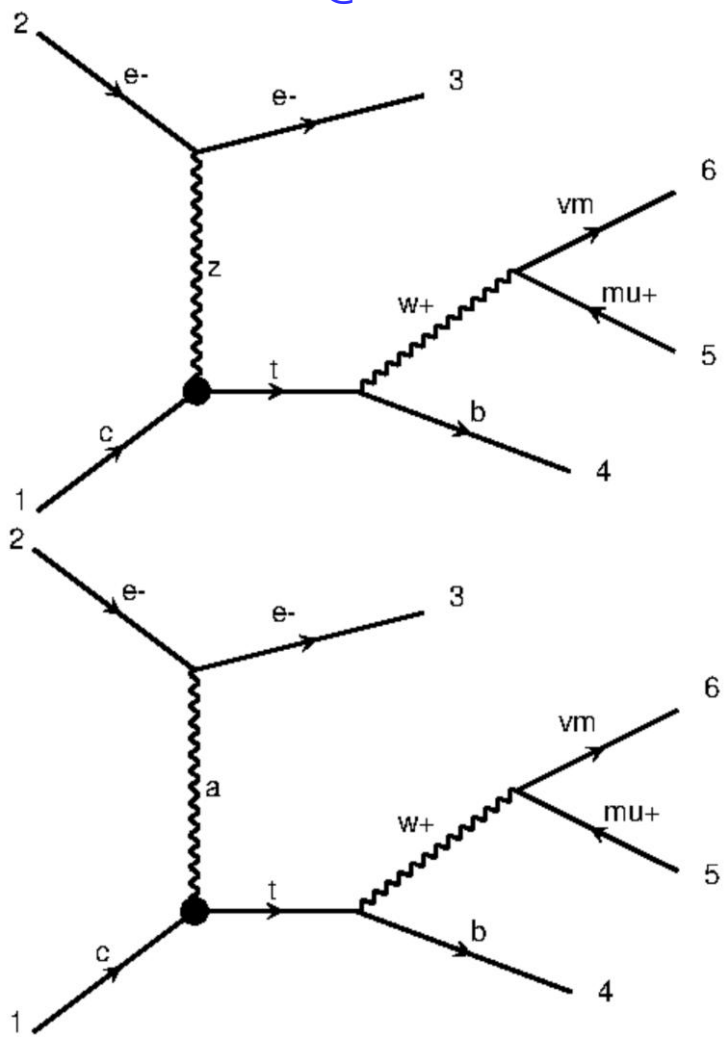
  - $p_T > 10$ GeV and $|\eta| < 5$ for **light jets**, **b-jets**, **leptons**
  - $\not{E}_T > 10$ GeV
  - $\Delta R_{ij} > 0.4$ between jets/leptons/photons

- Key analysis-level object requirement

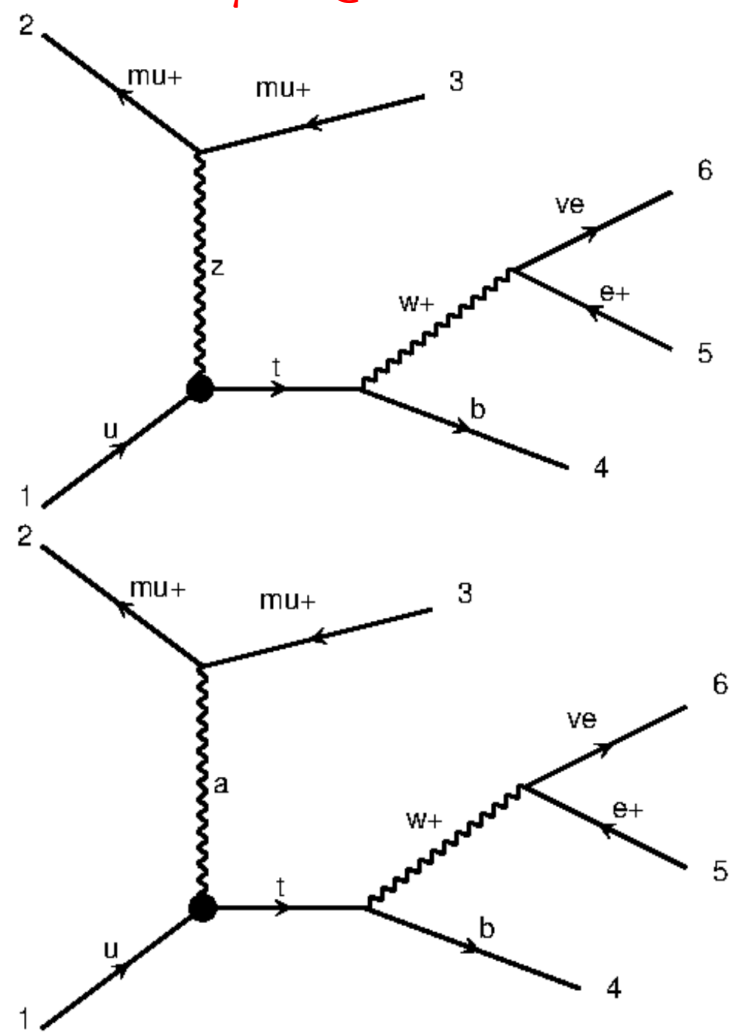
  - One scattered electron/muon
  - One b-jet
  - One lepton ($l^{\pm}$)

Top-FCNCs : Semi-leptonic decay channel

LeHC@1.2 TeV

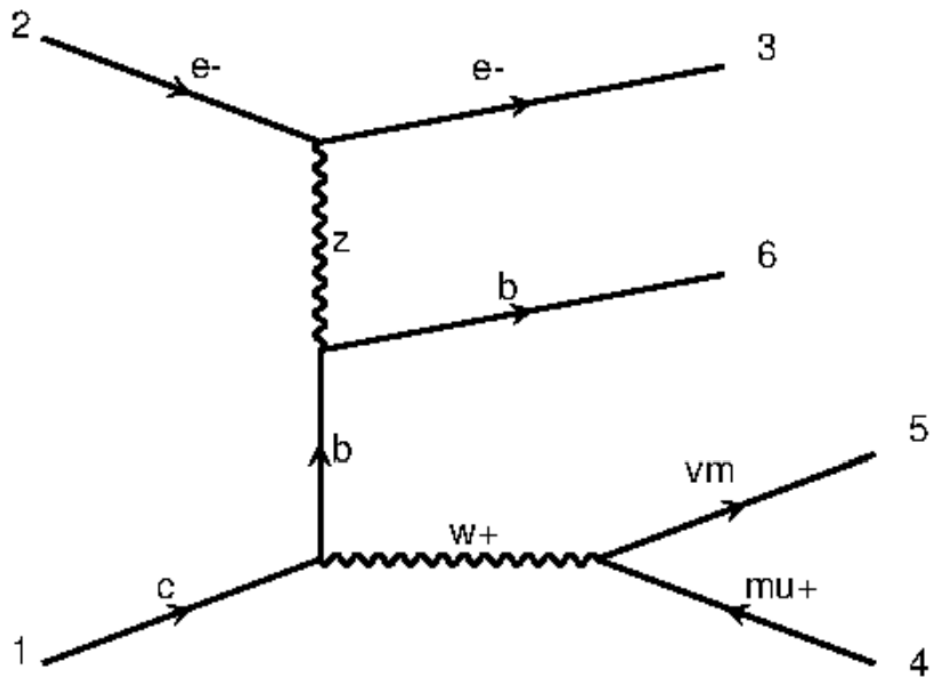


LμHC@3.7 TeV

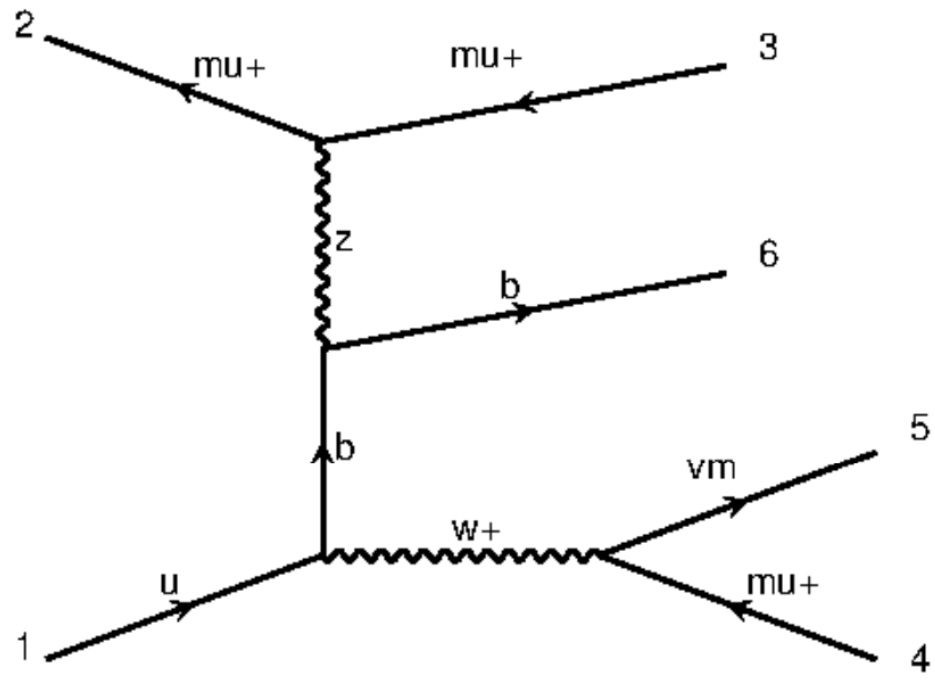


Background with the closest topology

LeHC@1.2 TeV



LμHC@3.7 TeV



List of SM backgrounds

- Charged-current (CC) backgrounds
- Neutral-current (NC) backgrounds
- Photoproduction backgrounds (γp initiated)

LHeC

(i) CC backgrounds

1. $e^- p \to e^- j W \to e^- j j j$
2. $e^- p \to e^- W j \to e^- j \ell \nu$
3. $e^- p \to \nu_e W b \to \nu_e \ell \nu b$

(ii) NC backgrounds

1. $e^- p \to e^- Z b \to e^- b b b$
2. $e^- p \to e^- Z b \to e^- b \ell^+ \ell^-$
3. $e^- p \to e^- Z b \to e^- b j j$

(iii) Photoproduction

1. $\gamma p \to \ell^+ \ell^- b$
2. $\gamma p \to \nu \bar{\nu} \ell^+ \ell^- b$

LHμC

(i) CC backgrounds

1. $\mu^+ p \to \mu^+ j W \to \mu^+ j j j$
2. $\mu^+ p \to \mu^+ W j \to \mu^+ j \ell^+ \nu_\ell$
3. $\mu^+ p \to \bar{\nu}_\mu W^+ b \to \bar{\nu}_\mu b \ell^+ \nu_\ell$

(ii) NC backgrounds

1. $\mu^+ p \to \mu^+ Z b \to \mu^+ b b b$
2. $\mu^+ p \to \mu^+ Z b \to \mu^+ b \ell^+ \ell^-$
3. $\mu^+ p \to \mu^+ Z b \to \mu^+ b j j$

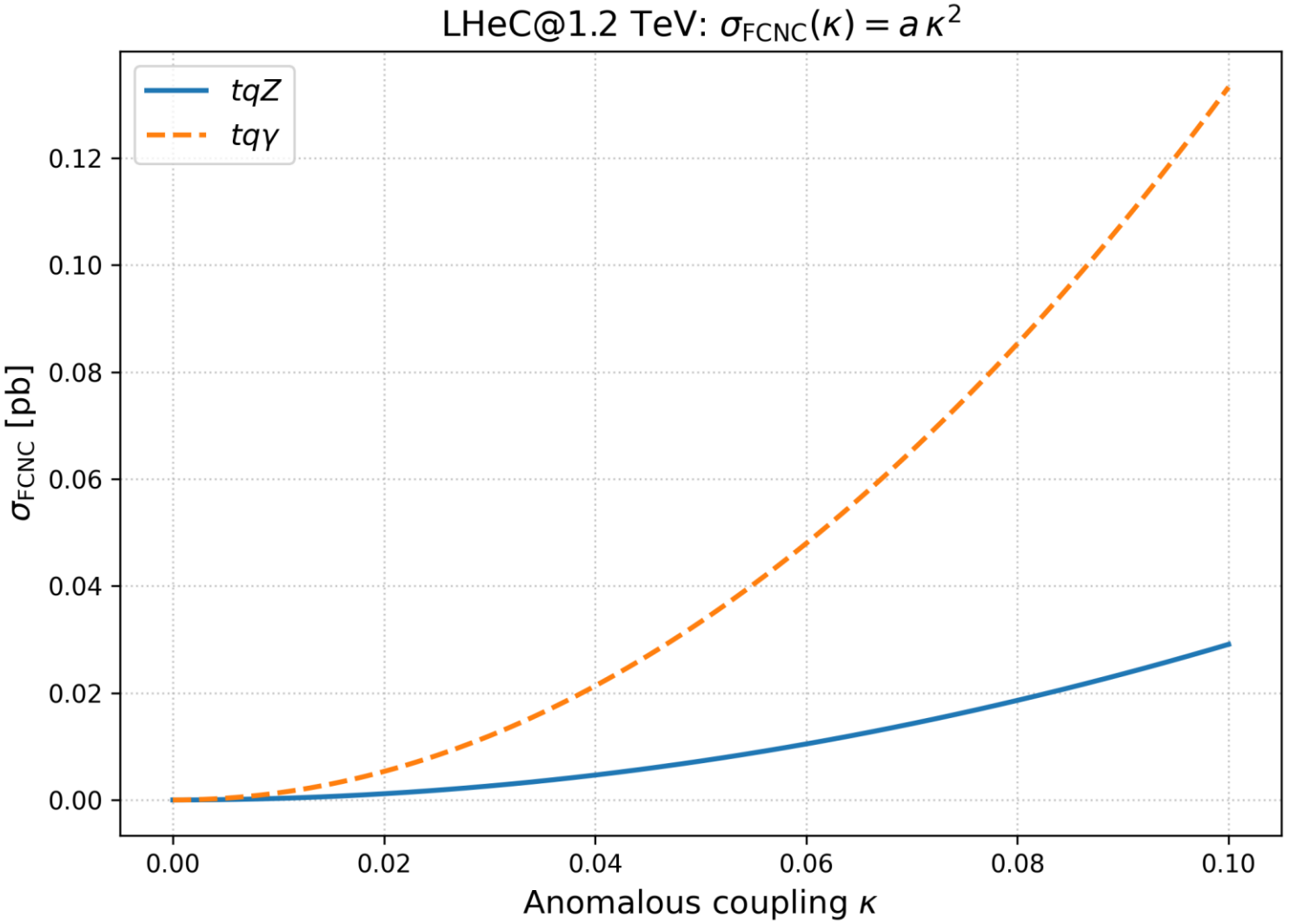
(iii) Photoproduction

1. $\gamma p \to \ell^+ \ell^- b$
2. $\gamma p \to \nu \bar{\nu} \ell^+ \ell^- b$

Sensitivity estimation: LHeC@1.2 TeV

- The cross sections induced by the tqZ and $tq\gamma$ operators exhibit the expected quadratic dependence on the coupling strength, $\sigma_{\mathrm{FCNC}} \propto \kappa^2$.

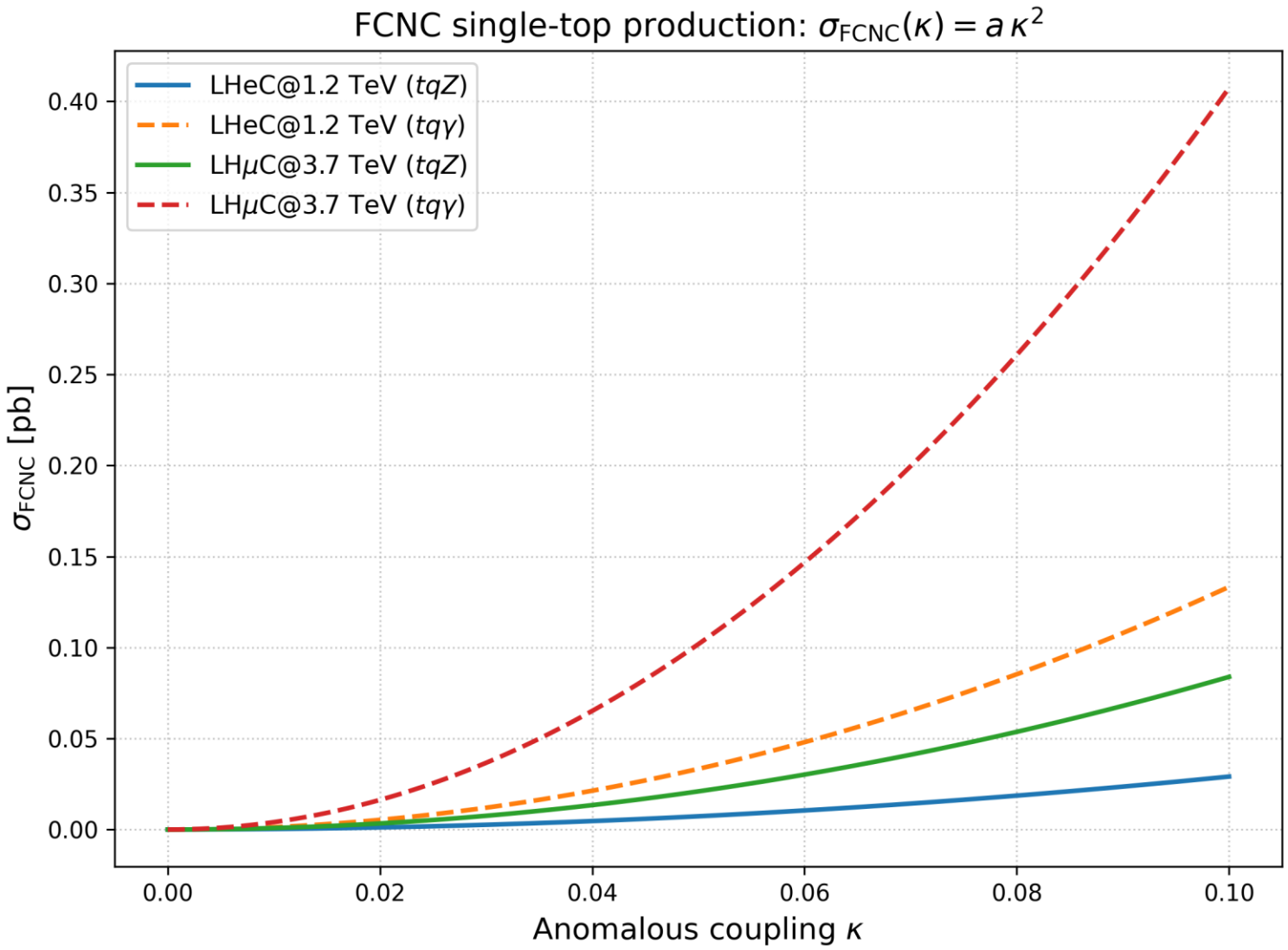
- More sensitivity to the $tq\gamma$ FCNC coupling.



FCNC single-top production cross sections at the LHeC@1.2 TeV as a function of the anomalous coupling $\boldsymbol{\kappa}$.

Sensitivity estimation: LHeC@1.2 TeV vs. LHμC@3.7 TeV

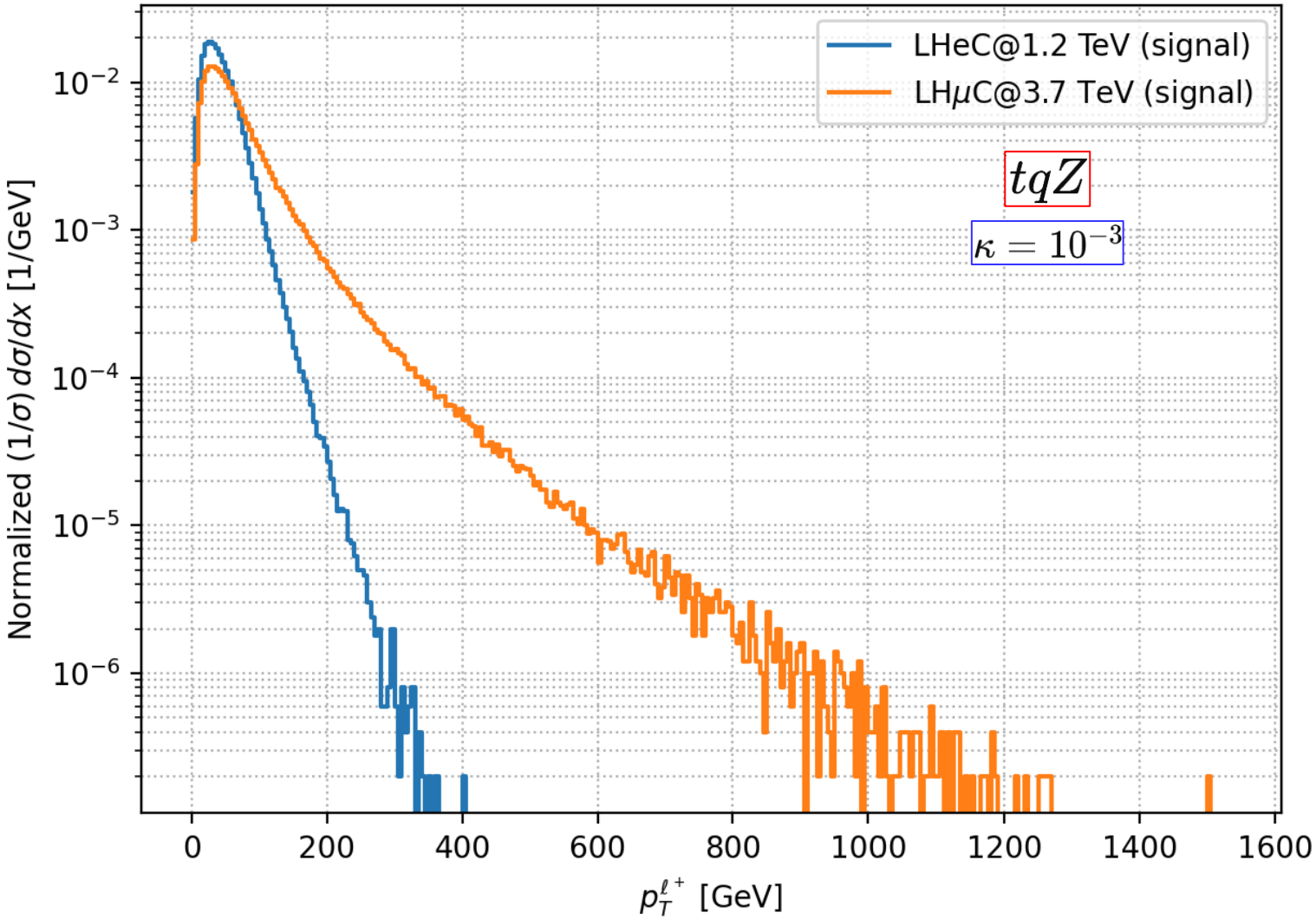
- The FCNC single-top production cross section exhibits a quadratic dependence on the anomalous coupling for both LHeC and LHμC configurations.
- Owing to the **higher center-of-mass energy**, the LHμC@3.7 TeV yields cross sections larger by approximately a factor of three compared to the LHeC@1.2 TeV, translating into a significantly improved sensitivity to FCNC couplings.



FCNC single-top production cross sections at the LHeC@1.2 TeV and LHμC@3.7 TeV as a function of the anomalous coupling $\boldsymbol{\kappa}$.

LHeC@1.2 TeV vs. LHμC@3.7 TeV

The LHμC exhibits a significantly harder $p_T^{\ell^+}$ spectrum for the lepton, with an extended high-p_T tail reaching the TeV scale, reflecting the higher center-of-mass energy and more boosted event kinematics compared to the LHeC.



The normalized differential distribution $(1/\sigma)\, d\sigma/dp_T^{\ell^+}$ for the top FCNC production at **LHeC@1.2 TeV** (blue) and **LHμC@3.7 TeV** (orange).

LHeC@1.2 TeV vs. LHμC@3.7 TeV

- **Kinematic reach:** LHμC at $\sqrt{s} = 3.7$ TeV produces substantially harder lepton p_T spectra, extending well beyond the LHeC kinematic limit.

- **Boosted top regime:** Single-top events at LHμC are more boosted, enhancing sensitivity to high-p_T observables and tail-based analyses.

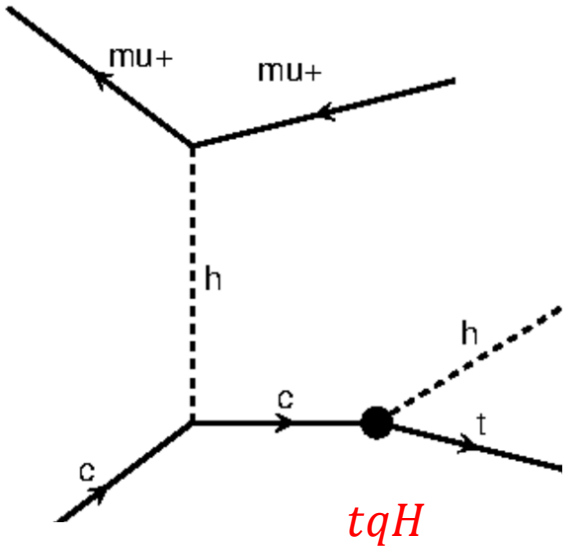
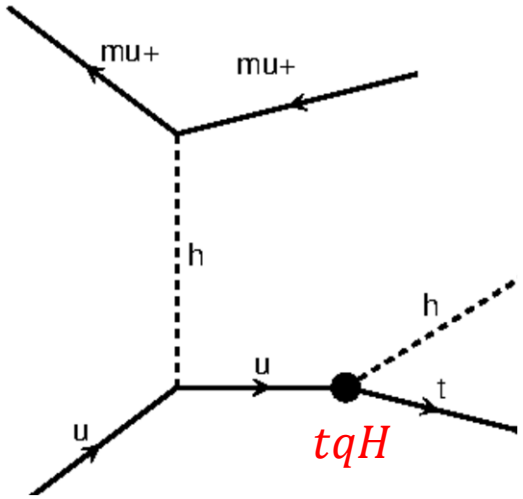
- **Signal-background separation:** The harder tails at LHμC enable more aggressive p_T cuts, where SM backgrounds fall faster, improving S/B.

- **Sensitivity to rare couplings:** Tail-dominated regions at LHμC provide stronger reach for FCNC and EFT operators compared to LHeC.

- **Complementarity:** LHeC offers cleaner moderate-energy kinematics and precision control, while LHμC excels in probing extreme kinematic regimes and rare effects.

Thank
you!

Top-FCNC couplings: tqH

generate p mu+ > mu+ t h j NP=1

$$p\,\mu^+ \to \mu^+\,t\,h\,j \qquad (\mathrm{FCNC},\ \mathrm{EFT},\ NP=1)$$